

Punnett Square Examples

1. Monohybrid: Pp × Pp (Mendel)

P = purple (dominant), p = white (recessive)

	P	p
P	PP	Pp
p	Pp	pp

Genotype: 1 PP : 2 Pp : 1 pp

Phenotype: 3 purple : 1 white (3:1)

2. Incomplete Dom: RW × RW

Snapdragons: R red + W white = pink heterozygo

	R	W
R	RR red	RW pink
W	RW pink	WW white

1 red : 2 pink : 1 white (1:2:1)

3 phenotypes because heterozygote is BLENDED

3. Dihybrid: TtYy × TtYy (classic 9:3:3:1)

T = tall, t = short · Y = yellow, y = green

	TY	Ty	tY	ty
TY	TTYy	TTYy	TtYY	TtYy
Ty	TTYy	TTyy	TtYy	Ttyy
tY	TtYY	TtYy	ttYY	ttYy
ty	TtYy	Ttyy	ttYy	ttyy

9 tall yellow (T_Y_)

3 tall green (T_yy)

3 short yellow (ttY_)

1 short green (ttyy)

Memorize: 9:3:3:1

Always = result of heterozygous
× heterozygous dihybrid